

Section II

present chapter by the following rules, which embody in some sense 'definitions' of ' $\wedge$ ', ' $\vee$ ', ' $\leftrightarrow$ ' in terms of ' $\sim$ ' and ' $\rightarrow$ ':

$$\begin{array}{rcl} \frac{(\phi \wedge \psi)}{\sim(\phi \rightarrow \sim\psi)} & \frac{\sim(\phi \rightarrow \sim\psi)}{(\phi \wedge \psi)} & \\ \frac{\phi \vee \psi}{(\sim\phi \rightarrow \psi)} & \frac{(\sim\phi \rightarrow \psi)}{(\phi \vee \psi)} & \\ \frac{(\phi \leftrightarrow \psi)}{((\phi \rightarrow \psi) \wedge (\psi \rightarrow \phi))} & \frac{((\phi \rightarrow \psi) \wedge (\psi \rightarrow \phi))}{(\phi \leftrightarrow \psi)} & \end{array}$$

The observation made at the end of section 5, that there is no automatic procedure which, in the absence of annotations, will permit a decision as to the correctness of an abbreviated derivation, holds in the sentential calculus only when infinitely many premises are permitted. It can be shown, however, that even when the class of premises, though allowed to be infinite, is required to be decidable (that is, loosely speaking, such that it can automatically be decided of any given sentence whether it is a member of the class), it is not generally possible to formulate an automatic test for a correct abbreviated derivation from that class of premises without recourse to annotations. In abbreviated derivations of the quantifier calculus (that is, the system of chapters III and IV), annotations are indispensable even when only finitely many premises are involved.

The story of exercise 77 is slightly modified from the original as given by Miguel de Cervantes (1615); it appears in Church [3]. The story of exercise 78 is due to Keisler and is a slight modification of its appearance in Kleene [2]. The story of exercise 79 is an old logical puzzle; it appears in Kemeny [1] and is discussed in Gardner [1].

II. Appendix: list of theorems of chapters I and II.

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|----------------|---|
| T ₁ | $P \rightarrow P$ |
| T ₂ | $Q \rightarrow (P \rightarrow Q)$ |
| T ₃ | $P \rightarrow ([P \rightarrow Q] \rightarrow Q)$ |
| T ₄ | $(P \rightarrow Q) \rightarrow ([Q \rightarrow R] \rightarrow [P \rightarrow R])$ |
| T ₅ | $(Q \rightarrow R) \rightarrow ([P \rightarrow Q] \rightarrow [P \rightarrow R])$ |
| T ₆ | $(P \rightarrow [Q \rightarrow R]) \rightarrow ([P \rightarrow Q] \rightarrow [P \rightarrow R])$ |
| T ₇ | $([P \rightarrow Q] \rightarrow [P \rightarrow R]) \rightarrow (P \rightarrow [Q \rightarrow R])$ |
| T ₈ | $(P \rightarrow [Q \rightarrow R]) \rightarrow (Q \rightarrow [P \rightarrow R])$ |

T ₉	$(P \rightarrow [P \rightarrow Q]) \rightarrow (P \rightarrow Q)$
T ₁₀	$([P \rightarrow Q] \rightarrow Q) \rightarrow ([Q \rightarrow P] \rightarrow P)$
T ₁₁	$\sim \sim P \rightarrow P$
T ₁₂	$P \rightarrow \sim \sim P$
T ₁₃	$(P \rightarrow Q) \rightarrow (\sim Q \rightarrow \sim P)$
✓ T ₁₄	$(P \rightarrow \sim Q) \rightarrow (Q \rightarrow \sim P)$
✓ T ₁₅	$(\sim P \rightarrow Q) \rightarrow (\sim Q \rightarrow P)$
✓ T ₁₆	$(\sim P \rightarrow \sim Q) \rightarrow (Q \rightarrow P)$
T ₁₇	$P \rightarrow (\sim P \rightarrow Q)$
T ₁₈	$\sim P \rightarrow (P \rightarrow Q)$
T ₁₉	$(\sim P \rightarrow P) \rightarrow P$
T ₂₀	$(P \rightarrow \sim P) \rightarrow \sim P$
✓ T ₂₁	$\sim (P \rightarrow Q) \rightarrow P$
✓ T ₂₂	$\sim (P \rightarrow Q) \rightarrow \sim Q$
✓ T ₂₃	$([P \rightarrow Q] \rightarrow P) \rightarrow P$
T ₂₄	$P \wedge Q \leftrightarrow Q \wedge P$
T ₂₅	$P \wedge (Q \wedge R) \leftrightarrow (P \wedge Q) \wedge R$
T ₂₆	$(P \rightarrow Q) \wedge (Q \rightarrow R) \rightarrow (P \rightarrow R)$
T ₂₇	$(P \wedge Q \rightarrow R) \leftrightarrow (P \rightarrow [Q \rightarrow R])$
T ₂₈	$(P \wedge Q \rightarrow R) \leftrightarrow (P \wedge \sim R \rightarrow \sim Q)$
T ₂₉	$(P \rightarrow Q \wedge R) \leftrightarrow (P \rightarrow Q) \wedge (P \rightarrow R)$
T ₃₀	$(P \rightarrow Q) \rightarrow (R \wedge P \rightarrow R \wedge Q)$
T ₃₁	$(P \rightarrow Q) \rightarrow (P \wedge R \rightarrow Q \wedge R)$
T ₃₂	$(P \rightarrow R) \wedge (Q \rightarrow S) \rightarrow (P \wedge Q \rightarrow R \wedge S)$
T ₃₃	$(P \rightarrow Q) \wedge (\sim P \rightarrow Q) \rightarrow Q$
T ₃₄	$(P \rightarrow Q) \wedge (P \rightarrow \sim Q) \rightarrow \sim P$
T ₃₅	$(\sim P \rightarrow R) \wedge (Q \rightarrow R) \leftrightarrow ([P \rightarrow Q] \rightarrow R)$
T ₃₆	$\sim (P \wedge \sim P)$
T ₃₇	$(P \rightarrow Q) \leftrightarrow \sim (P \wedge \sim Q)$

T96	$(P \leftrightarrow Q) \leftrightarrow (\sim P \leftrightarrow \sim Q)$
T97	$(P \leftrightarrow R) \wedge (Q \leftrightarrow S) \rightarrow ([P \rightarrow Q] \leftrightarrow [R \rightarrow S])$
T98	$(P \leftrightarrow R) \wedge (Q \leftrightarrow S) \rightarrow (P \wedge Q \leftrightarrow R \wedge S)$
T99	$(P \leftrightarrow R) \wedge (Q \leftrightarrow S) \rightarrow (P \vee Q \leftrightarrow R \vee S)$
T100	$(P \leftrightarrow R) \wedge (Q \leftrightarrow S) \rightarrow ([P \leftrightarrow Q] \leftrightarrow [R \leftrightarrow S])$
T101	$(Q \leftrightarrow S) \rightarrow ([P \rightarrow Q] \leftrightarrow [P \rightarrow S]) \wedge ([Q \rightarrow P] \leftrightarrow [S \rightarrow P])$
T102	$(Q \leftrightarrow S) \rightarrow (P \wedge Q \leftrightarrow P \wedge S)$
T103	$(Q \leftrightarrow S) \rightarrow (P \vee Q \leftrightarrow P \vee S)$
T104	$(Q \leftrightarrow S) \rightarrow ([P \leftrightarrow Q] \leftrightarrow [P \leftrightarrow S])$
T105	$P \wedge (Q \leftrightarrow R) \rightarrow (P \wedge Q \leftrightarrow R)$
T106	$(P \rightarrow [Q \rightarrow R]) \leftrightarrow ([P \rightarrow Q] \rightarrow [P \rightarrow R])$
T107	$(P \rightarrow [Q \rightarrow R]) \leftrightarrow (Q \rightarrow [P \rightarrow R])$
T108	$(P \rightarrow [P \rightarrow Q]) \leftrightarrow (P \rightarrow Q)$
T109	$([P \rightarrow Q] \rightarrow Q) \leftrightarrow ([Q \rightarrow P] \rightarrow P)$
T110	$P \leftrightarrow \sim \sim P$
T111	$(P \rightarrow Q) \leftrightarrow (\sim Q \rightarrow \sim P)$
T112	$(P \rightarrow \sim Q) \leftrightarrow (Q \rightarrow \sim P)$
T113	$(\sim P \rightarrow Q) \leftrightarrow (\sim Q \rightarrow P)$
T114	$(\sim P \rightarrow P) \leftrightarrow P$
T115	$(P \rightarrow \sim P) \leftrightarrow \sim P$
T116	$(P \wedge Q) \vee (R \wedge S) \leftrightarrow (P \vee R) \wedge (P \vee S) \wedge (Q \vee R) \wedge (Q \vee S)$
T117	$(P \vee Q) \wedge (R \vee S) \leftrightarrow (P \wedge R) \vee (P \wedge S) \vee (Q \wedge R) \vee (Q \wedge S)$
T118	$(P \rightarrow Q) \wedge (R \rightarrow S) \leftrightarrow$ $(\sim P \wedge \sim R) \vee (\sim P \wedge S) \vee (Q \wedge \sim R) \vee (Q \wedge S)$
T119	$(P \vee \sim P) \wedge Q \leftrightarrow Q$
T120	$(P \wedge \sim P) \vee Q \leftrightarrow Q$
T121	$P \vee (\sim P \wedge Q) \leftrightarrow P \vee Q$
T122	$P \wedge (\sim P \vee Q) \leftrightarrow P \wedge Q$
T123	$P \leftrightarrow P \vee (P \wedge Q)$

112 . 'And', 'Or', 'If' and 'Only If'

$$T_{124} \quad P \leftrightarrow P \wedge (P \vee Q)$$

$$T_{125} \quad (P \rightarrow Q \wedge R) \rightarrow (P \wedge Q \leftrightarrow P \wedge R)$$

12. Appendix: solutions to selected exercises.

Section 1

No. 3 is a symbolic sentence. Nos. 4 and 5 are not; the former fails to be a symbolic sentence because ' $(P \vee Q \rightarrow R)$ ' lacks a pair of parentheses, and the latter because the ' $\rightarrow$ ' fails to bring with it a pair of parentheses.

$$7. ((Q \vee R) \rightarrow ((P \wedge R) \rightarrow (Q \leftrightarrow (R \vee P))))$$

$$10. \sim((P \vee (Q \wedge R)) \rightarrow (((P \vee Q) \wedge (P \vee R)) \leftrightarrow (P \wedge Q)))$$

Section 2

11. Neither Alice nor Mary will dance with Alfred unless he will improve his deportment, but Alfred will improve his deportment if and only if both Alice and Mary will dance with him.

To obtain symbolizations of the remaining sentences of these exercises, we first employ parentheses together with sentence letters and negation signs to obtain a reading, and follow this with a symbolization.

13. Assuming that either P or $\sim Q$, R only if S; $P \vee \sim Q \rightarrow (R \rightarrow S)$

14. Unless P, R if S; $P \vee (S \rightarrow R)$

15. In this sentence 'unless' is to be understood in the exclusive sense. Thus, 'P unless Q' may pass into ' $(P \vee Q) \wedge \sim(P \wedge Q)$ '.

16. Assuming that P, Q; but if R, neither Q nor S; $(P \rightarrow Q) \wedge [R \rightarrow \sim(Q \vee S)]$

17. In the sentence 'Alfred and Mary and playing dice together' the word 'and' is not used as a phrase of connection; however, the word 'or' in the sentence 'Mary throws 7 or 11' has such a use. Thus, in symbolizing No. 17 we may not use the sentence letters 'P' and 'Q', but we may use either 'V' or both 'W' and 'X'. On the basis of the given scheme, we can obtain two symbolizations of No. 17:

If R, S, and T, then U if and only if V; $(R \wedge S) \wedge T \rightarrow (U \leftrightarrow V)$

If R, S, and T, then U if and only if (W or X); $(R \wedge S) \wedge T \rightarrow (U \leftrightarrow W \vee X)$

18. If P, then either Q or, provided that R, S; $P \rightarrow Q \vee (R \rightarrow S)$

20. If (either P or Q), then (neither R nor [S and T]) unless (U and V); $(P \vee Q) \rightarrow \sim(R \vee [S \wedge T]) \vee (U \wedge V)$

If (either P or Q) then (neither R nor [S and T]), unless (U and V); $[(P \vee Q) \rightarrow \sim(R \vee [S \wedge T])] \vee (U \wedge V)$
(Other alternative readings are less plausible.)

21. (Neither P nor Q) if either R or S; $R \vee S \rightarrow \sim(P \vee Q)$

Neither P nor (Q if either R or S); $\sim(P \vee [R \vee S \rightarrow Q])$

Neither (P if either R or S) nor (Q if either R or S); $\sim[(R \vee S \rightarrow P) \vee (R \vee S \rightarrow Q)]$